AUTOSAR High-Level Design (HLD) Specification

System: Advanced Driver Assistance System (ADAS) ECU

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Classification: CONFIDENTIAL

1. Introduction

This document specifies the High-Level Design for the ADAS Electronic Control Unit (ECU) based on the AUTOSAR Classic Platform. It defines the Software Components (SWCs), the interfaces over the Runtime Environment (RTE), and the dependencies on the Basic Software (BSW).

1.1 Scope

The scope covers the application layer components for Adaptive Cruise Control (ACC) and Lane Keeping Assist (LKA). It includes integration with vehicle networks (CAN-FD, Automotive Ethernet) and cybersecurity modules (SecOC).

2. Software Component (SWC) Architecture

2.1 Component: ACC_Controller

The ACC_Controller SWC is responsible for calculating the target acceleration and braking force to maintain a safe distance from the preceding vehicle.

Interface Name	Port Type	Data Element	Description
RadarData_Rx	RPort	VehicleDistance	Distance to leading vehicle (cm)
Speedometer_Rx	RPort	CurrentSpeed	Ego vehicle speed (kph)
ThrottleRequest_Tx	PPort	TargetAcceleration	Acceleration request to Engine ECU
BrakeRequest_Tx	PPort	TargetDeceleration	Braking request to Brake ECU

2.2 Component: LKA_Controller

The LKA_Controller processes camera images to detect lane markings and provides steering torque requests to keep the vehicle centered.

Interface Name	Port Type	Data Element	Description
CameraVision_Rx	RPort	LaneDeviation	Lateral deviation from lane center (cm)
SteeringActuator_Tx	PPort	TorqueRequest	Requested steering torque (Nm)

3. Basic Software (BSW) Dependencies

3.1 Diagnostic Communication Manager (DCM)

The ADAS ECU supports UDS (ISO 14229) diagnostics. The DCM must handle the following services: 0x22 (ReadDataByIdentifier) for fault reading, 0x19 (ReadDTCInformation), and 0x27 (SecurityAccess).

3.2 Non-Volatile Memory Manager (NvM)

Calibration data for the ACC_Controller (e.g., driver following distance preference) shall be stored in block NVM_BLOCK_ACC_PREFS. The block requires CRC32 validation and redundant storage.

4. Functional Safety & Cybersecurity

4.1 HARA & ASIL Classification

According to the Hazard Analysis and Risk Assessment (HARA), the ACC unintended acceleration hazard is classified as ASIL D. Therefore, the ThrottleRequest_Tx signal must use End-to-End (E2E) protection Profile 2 to ensure data integrity over the CAN bus.

4.2 Cybersecurity (ISO 21434)

To prevent spoofing attacks on the BrakeRequest_Tx interface, Secure Onboard Communication (SecOC) shall be enabled. A 64-bit Freshness Value and a CMAC truncation to 24 bits will be used to authenticate the braking messages.